

# Dynamic Capabilities & Risk Management

(Chapter 7 and Chapter 8)

**Yariv Taran**

IKE Research Group

Aalborg University Business School

[yariv@business.aau.dk](mailto:yariv@business.aau.dk)



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# Factors that drives company's performance

- **Industry-specific factors:**
  - Profitable/declining industries affect performance.
  - Industrial rivalry, power of suppliers/customers, and threats from new entrance/substitute products.
- **Company-specific factors (unique attributes)**
  - Company effects are a major contributor to performance:
    - it's not just industry context-dependent that affects performance.
    - important for understanding what are these “unique attributes” sources and their link to a firm's competitive advantage.
  - **Firm unique attributes account for most explainable performance effects**, in comparison to e.g. industry-specific factors (e.g. Hawawini et al. 2003; McGahan & Porter 1997; Rumelt 1991; Schmalensee 1985; Verdin, 2000).

# Sustained competitive advantage

- When a company's performance advantage is sustained over time, it can be said that it has a **sustainable** competitive advantage (e.g. Apple; NIKE; Amazon; Facebook; P&G; Intel...).
- Key important questions:
  - **Heterogeneity** – **What** are the **attributes needed** by the firm in order to make differences in performance (i.e., differentiate), and **how** a company **can acquire and develop** these attributes?
  - **Sustaining** – Once a company acquires a competitive advantage, how can it be **made to last**?

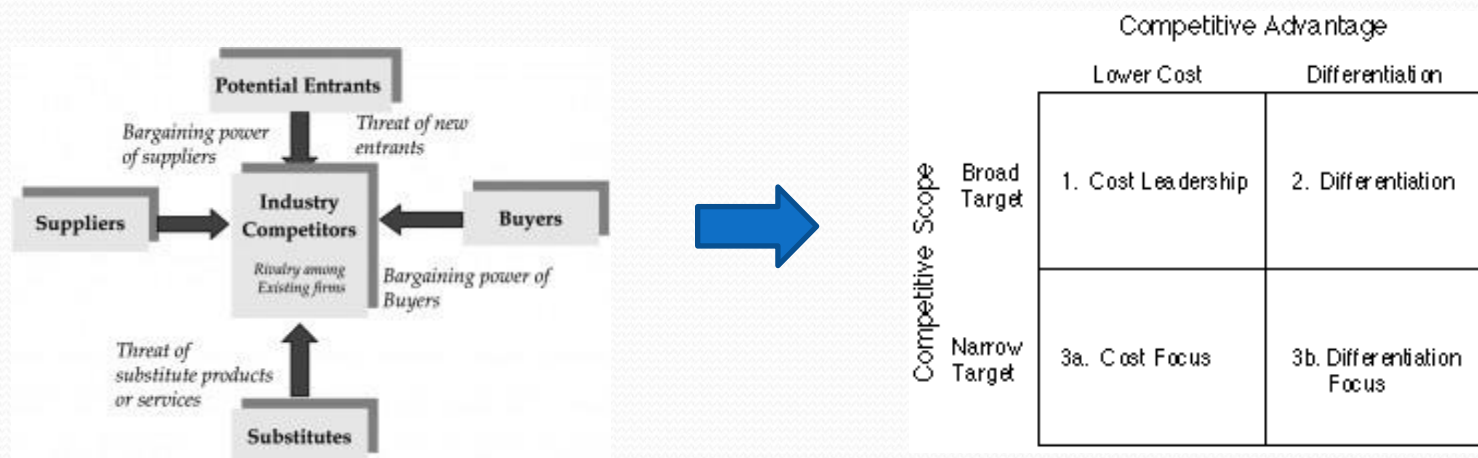
# What are the sources of "sustainable" competitive advantage?

- Main frameworks/paradigms (Teece et al. 1997):
  1. Porter's competitive advantage
  2. Strategic-conflict approaches
  3. Resource-based view
  4. Dynamic capabilities

..let's look at each of them one at a time:

# 1. Competitive advantage – Porter view

- Michael Porter (1980's) – argued that competitive advantage can be achieved when firms develop a **differential advantage** over their competitors. Here, competitive advantage is **discussed from a competitive perspective** (i.e. outside-in).
- Emphasizes the actions a firm can take to create **differentiated positioning** against competitive forces - **lead to three/four generic strategies for achieving above-average performance in an industry: cost leadership, differentiation, and focus.**



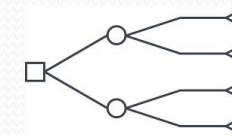
## 2. Competitive advantage - Strategic conflict

- View **competitive outcomes** as a function of the effectiveness with which **firms keep their rivals off balance through strategic investments, pricing strategies, signaling, and control of information** (Teece et al. 1997).
- Strategic conflict theory **had no claim to represent realistic conflicts.** **Mainly concerned with “rational conflict” or “threat”** as part of a bargaining procedure between parties who were supposed to be partners and/or adversaries.

# Strategic conflict theory - examples

- **Symmetric/Asymmetric game** - a game where the payoffs for playing a particular strategy **depend only on the other strategies employed**, not on who is playing them (e.g. Prisoner's dilemma..)
- **Zero-sum** (or even Non-zero-sum) - one person's gains result in losses for the other participants (Poker, Chess).
- **Perfect/imperfect information** (e.g. decision tree); Combinatorial games; Infinitely long games...

| YOU                |   | YOUR ACCOMPLICE |   | YOU     |         | YOUR ACCOMPLICE                            |  |
|--------------------|---|-----------------|---|---------|---------|--------------------------------------------|--|
| If you stay silent | 😊 | 😊               | 😊 | FREE    | FREE    | You might go free or you might get 5 years |  |
|                    | 😞 | 😞               | 😞 | 5 years | 2 years |                                            |  |
| If you confess     | 😞 | 😊               | 😊 | 2 years | 5 years | You can only ever get 2 years              |  |
|                    | 😞 | 😞               | 😞 | 2 years | 2 years |                                            |  |



- “Both the **Competitive Forces** (Porter) and the **Strategic Conflict** approaches appear to share the view that **rents (payments) flow from privileged product-market positions**”. (Teece et al. 1997).
- **Yet, traditional sources of competitive advantage are stripped away:**
  - Decreased cost and increased speed of information flow - **everybody knows everything these days**.
  - Dispersion in the sources of innovation, and value proposition’s **lifecycles and competencies**.
  - **Hypercompetition** - competition is high all the time..
  - Increase ease of organizing activities (e.g., outsourcing, offshoring), and access to new resources (e.g., less supply dependencies).
  - ..need for more **flexibility and agility**.



# 3. Competitive advantage - Resource-based view (RBV)

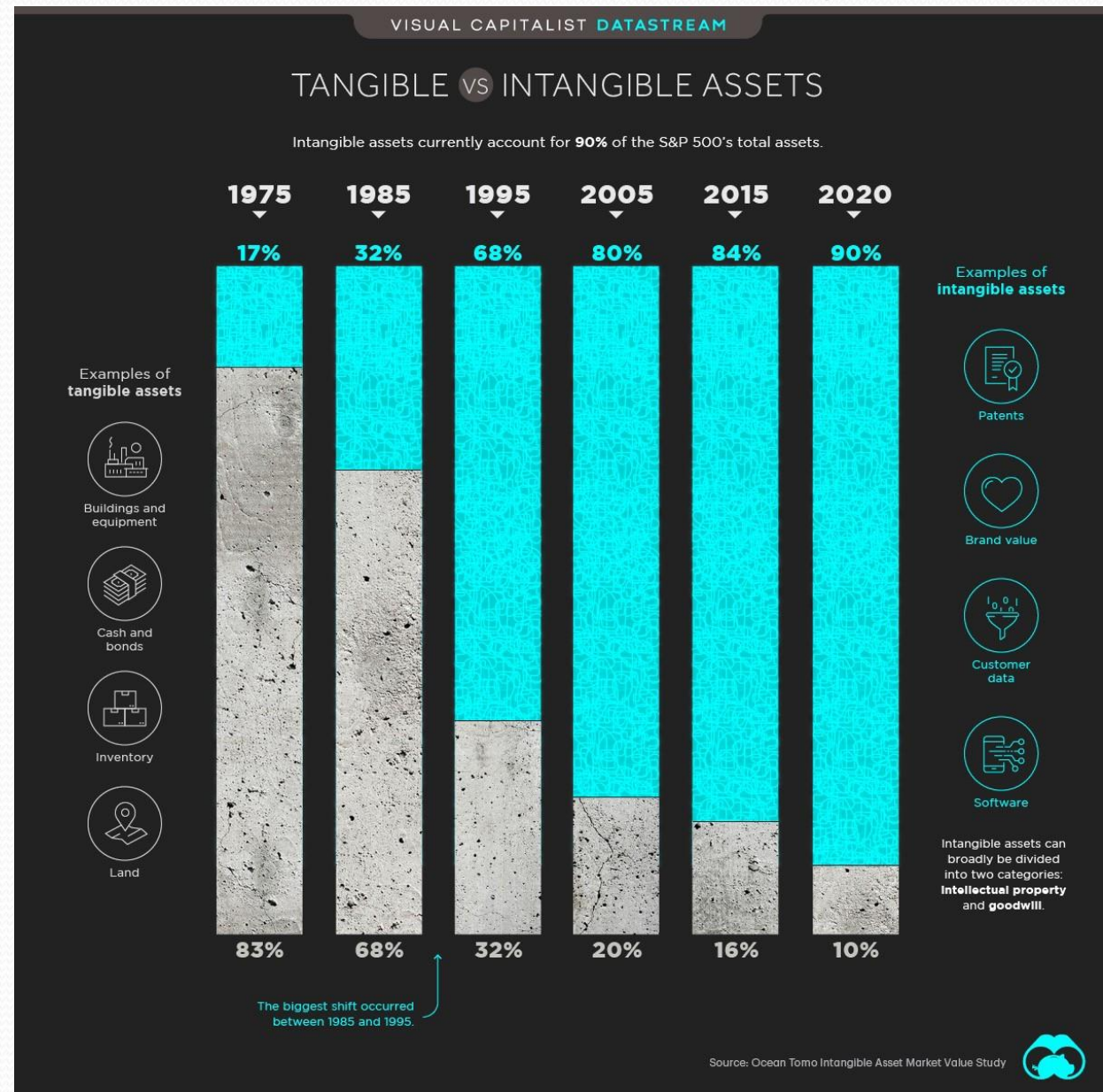
- Barney (1991) – Resource-based view (RBV). Competitive advantage is achieved (also) by looking into **internal aspects of the firm** – i.e., **looking inwards rather than (only) outwards**.
- Focus on strengthening internal **effectiveness** (i.e., doing the right things) and **efficiency** (i.e., doing things right), **assets, competencies & capabilities**.
  - Focus on strengthening **tangible** assets - Assets that are physical in nature, or which can be seen or touched.
  - Focus on strengthening **intangible** assets - Non-physical assets which are valuable and income-generating resources.

# Growing market value significance of intangible assets - when it comes to “sustainability”

“Over the last thirty years, the industrial economy has transitioned to become **the knowledge economy**. **Intellectual capital** has emerged as the **leading asset class**. In **1975**, more than 80% of the market’s value was composed of **tangible assets**: **factories, machines, and inventory**. In e.g., **2007**, less than 20% of the market’s value is composed of tangible assets, replaced by (legal) **intangible assets** and led by intellectual property (*patents, trade secrets, copyrights, trademarks, etc.*)”.

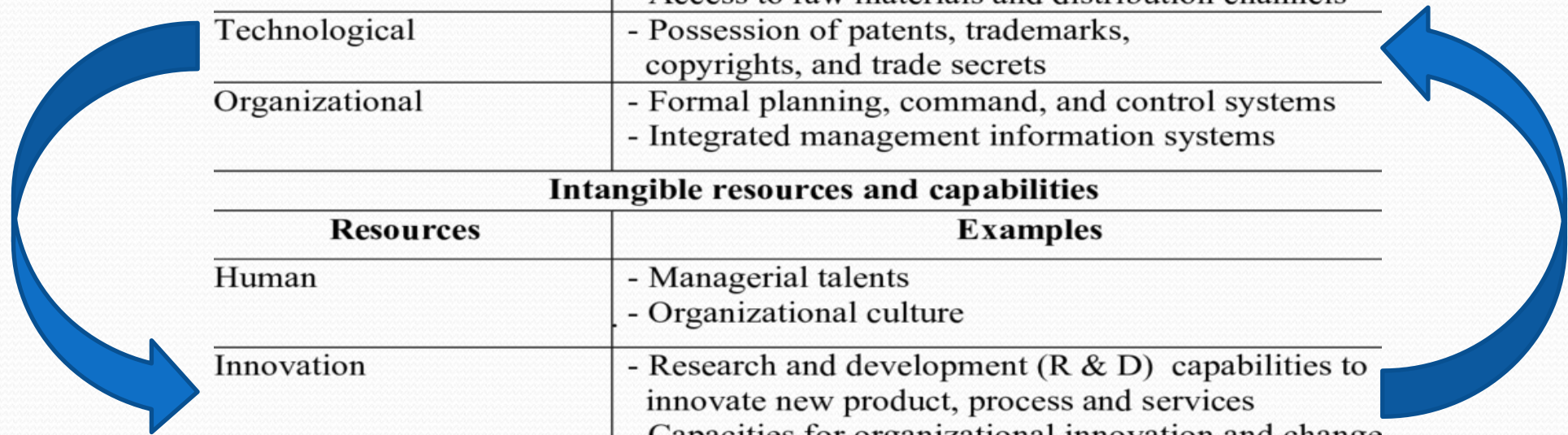
(Source: Ocean Tomo 300® Patent Index)

**The Soaring Value of Intangible Assets in the S&P 500** — source: <https://www.visualcapitalist.com>



# Resources types

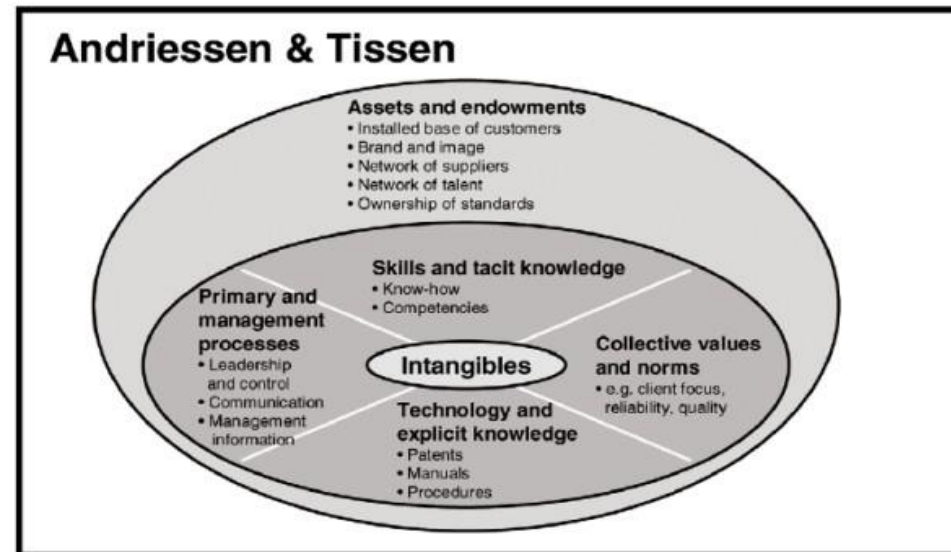
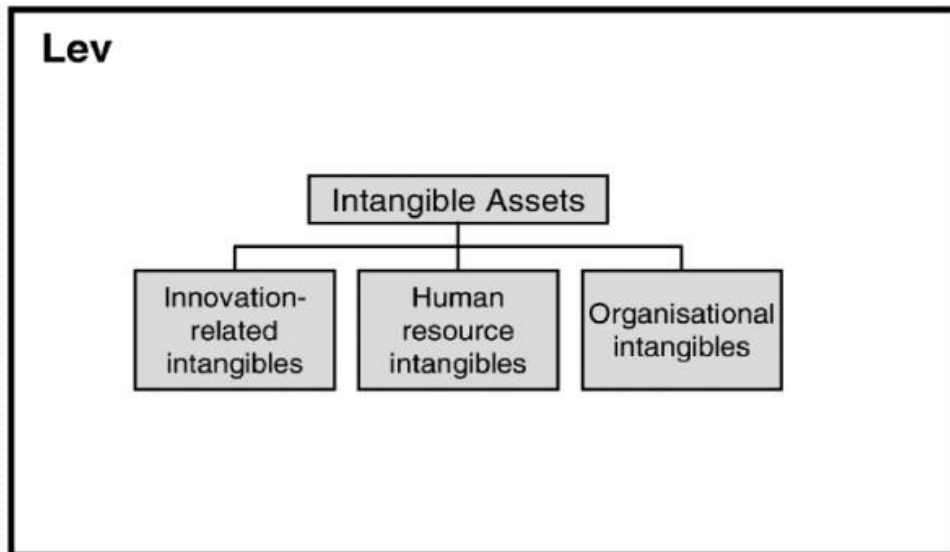
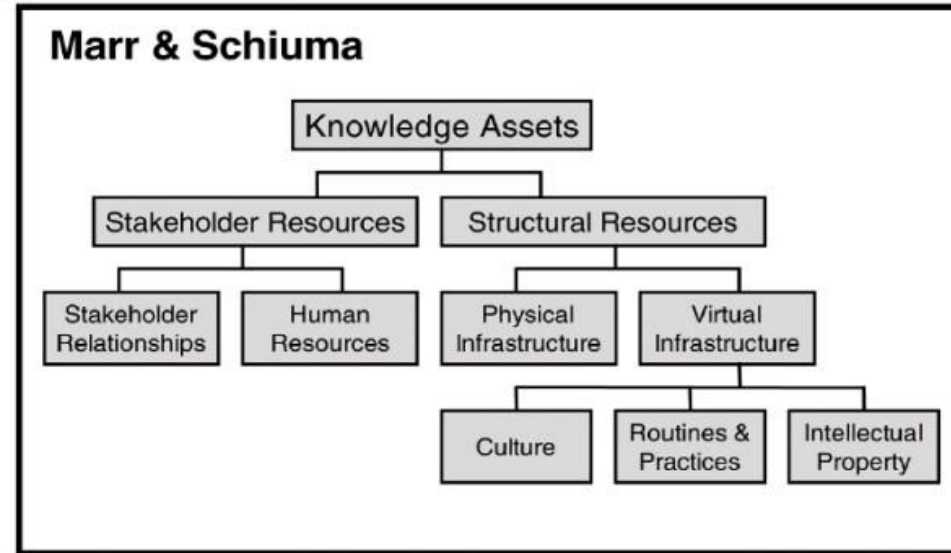
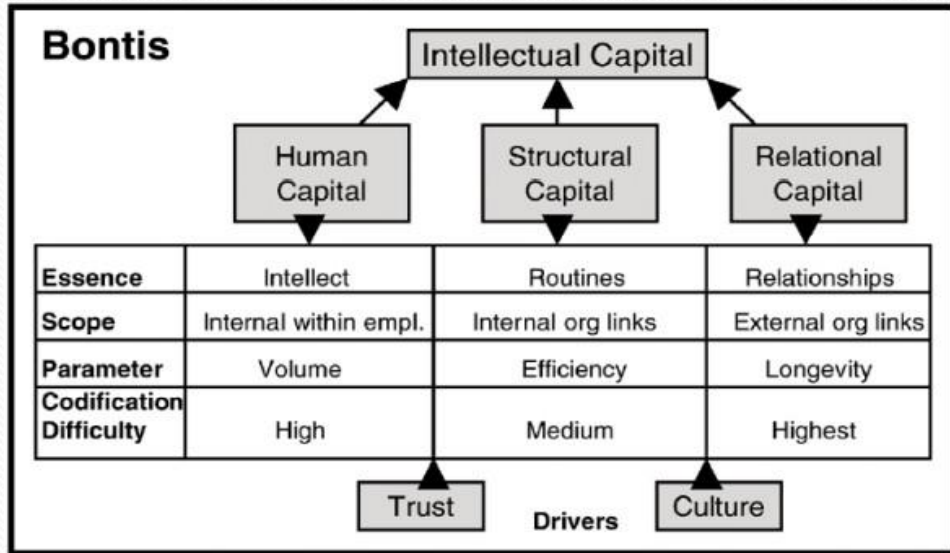
| Tangible resources and capabilities   |                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resources                             | Examples                                                                                                                                                                                                                                                                                                                          |
| Financial                             | <ul style="list-style-type: none"><li>- Ability to generate internal funds</li><li>- Ability to raise external capital</li></ul>                                                                                                                                                                                                  |
| Physical                              | <ul style="list-style-type: none"><li>- Location of plants, machines, offices, and their geographic locations</li><li>- Access to raw materials and distribution channels</li></ul>                                                                                                                                               |
| Technological                         | <ul style="list-style-type: none"><li>- Possession of patents, trademarks, copyrights, and trade secrets</li></ul>                                                                                                                                                                                                                |
| Organizational                        | <ul style="list-style-type: none"><li>- Formal planning, command, and control systems</li><li>- Integrated management information systems</li></ul>                                                                                                                                                                               |
| Intangible resources and capabilities |                                                                                                                                                                                                                                                                                                                                   |
| Resources                             | Examples                                                                                                                                                                                                                                                                                                                          |
| Human                                 | <ul style="list-style-type: none"><li>- Managerial talents</li><li>- Organizational culture</li></ul>                                                                                                                                                                                                                             |
| Innovation                            | <ul style="list-style-type: none"><li>- Research and development (R &amp; D) capabilities to innovate new product, process and services</li><li>- Capacities for organizational innovation and change</li></ul>                                                                                                                   |
| Reputational                          | <ul style="list-style-type: none"><li>- Perceptions of product quality, durability, and reliability among customers</li><li>- Successful product branding and positioning with satisfied and loyal customer base</li><li>- Reputation as a good employer</li><li>- Reputation as socially responsible corporate citizen</li></ul> |



Sources: Madhani, 2014 The Resource-Based View: Issues and Perspectives.

(Compiled by Pankaj M. 2010; Adapted from (1) J. Barney, 1991, Firm resources and sustained competitive advantage, Journal of Management, 17: 101; (2) R. Hall, 1992, The strategic analysis of intangible resources, Strategic Management Journal, 13: 135-144.)

# Challenges in operationalizing and measuring intangible assets – e.g.



Mar and Adams, 2004

# **Sustainability via resource-based view (RBV)**

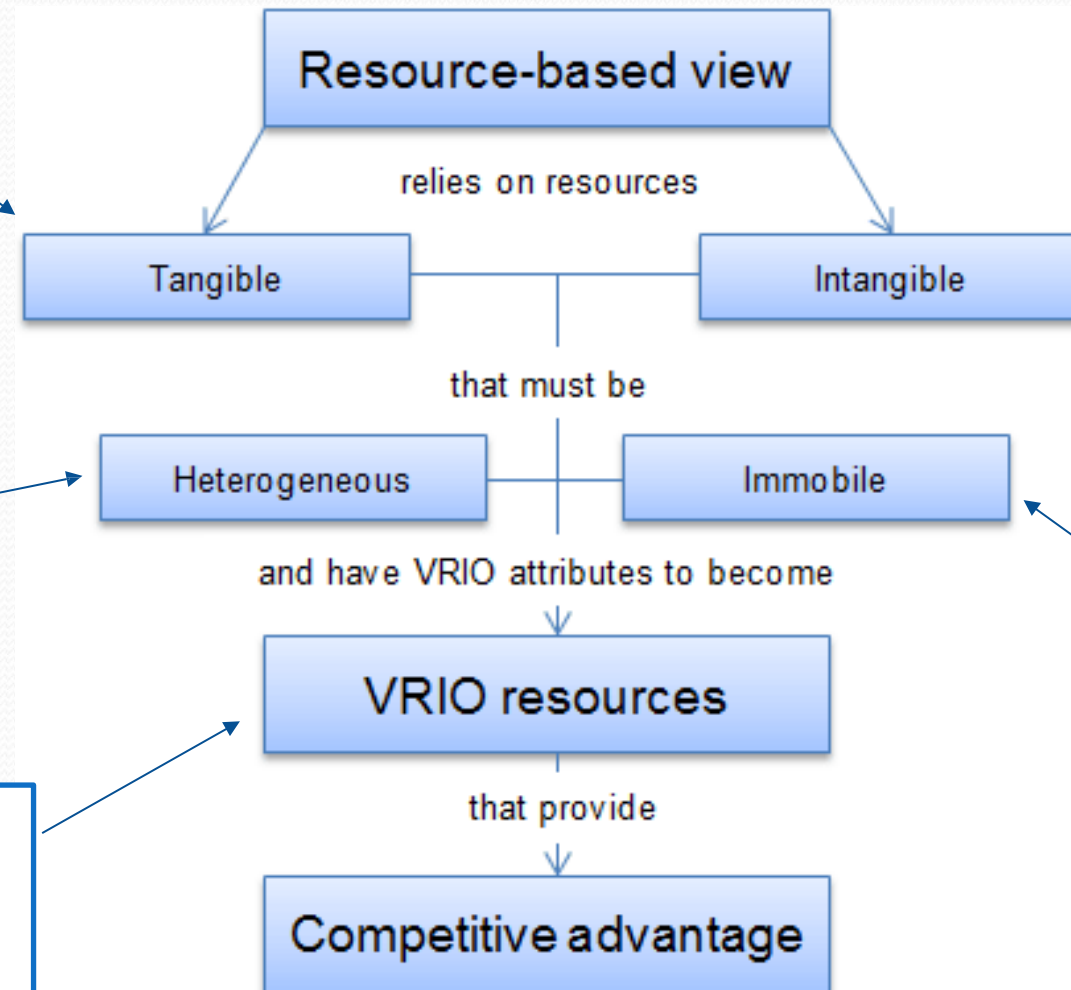
- A firm's **competitive position is defined by a bundle of unique resources and relationships'** (Rumelt, 1984), that are heterogeneously distributed among firms and are imperfectly mobile (Barney 1991).
  - **Imperfect mobility** – refers to the **non-tradability** and barriers to moving certain resources from one company to another.
- Here, **resources and products/services** are **two sides of the same coin**. Therefore, firms seek to acquire and develop unique sets of resources and capabilities to gain better competitive positioning in the marketplace (Jiang, 2014).

# Resource-based view (RBV)

**Physical resources** - can easily be bought in the market so they confer little advantage to the companies in the long run because rivals can soon acquire identical assets. E.g., land, vehicles, equipment, machinery, furniture, inventory, stock, bonds, and cash.

**Heterogeneous** - companies achieve competitive advantage by using **their different bundles of resources**. i.e., many resources and capabilities, well organized and harmonized throughout the organization.

**VRIO** – “Value, Rareness, Imitability, Organization”.  
**or**  
**VRIN** - “Valuable, Rare, Imitable and Non-substitutable.”



**Intangible resources** usually stay within a company and are the main source of sustainable competitive advantage.

**Immobile** - resources that do not move from company to company, at least in the short-run. Due to this immobility, **companies cannot replicate rivals' resources and implement the same strategies.**

# VRIO Analysis

Resources are **valuable** if they help organizations to **increase the value offered to the customers**. This is done by increasing differentiation and/or decreasing the costs of the production. The resources that cannot meet this condition, lead to competitive disadvantage.

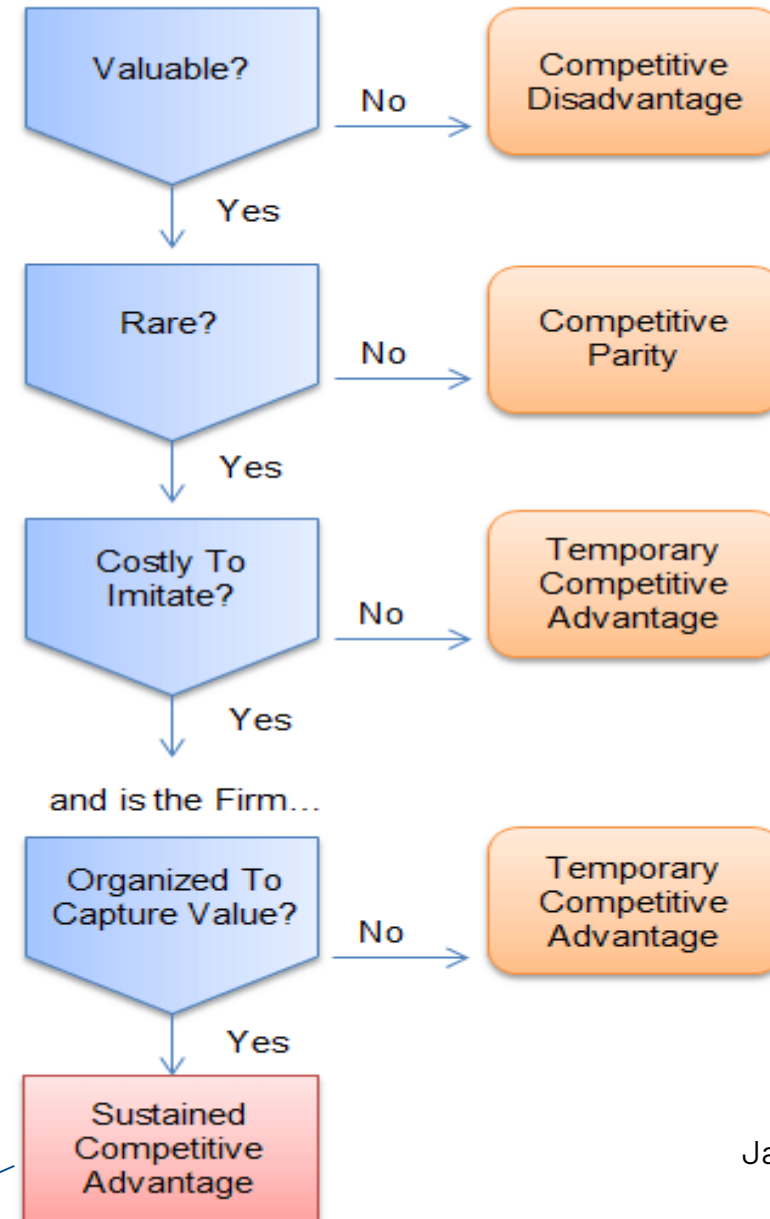
Resources that **can only be acquired by one or few companies** are considered **rare**. When more than few companies have the same resource or capability, it results in competitive parity.

A company that has valuable and rare resource can achieve at least temporary competitive advantage. However, the **resource must also be costly to imitate/substitute by rival's**, if a company wants to achieve sustained competitive advantage.

The resources themselves do not confer any advantage for a company if they're not **organized to capture the value from them**. Only the firm that is capable to exploit the valuable, rare and imitable resources can achieve sustained competitive advantage.

However, the framework does not explicitly address the long-term **durability** or changing **relevance** of resources in dynamic environments (cf. Dynamic Capabilities).

Is the Resource or Capability ...



Jay B. Barney

# Durability and relevancy limitations:

- Durability (or stability).
  - **Leaky bucket** – where the bucket represents the company, and the water is its resources and capabilities. Internal and external changing conditions slowly and steadily lead companies to keep losing competencies, offerings, skills, degrading resources, key employees leaving, etc.
- Relevance
  - In a constantly changing environment, the company's existing resources and capabilities may not be useful or relevant in the future. **Path dependencies can become an obstacle rather than a trigger to secure competitive advantage.**
  - The need here is to be able and **change organizational capabilities** when needed - i.e., dynamic capabilities.



# Limitations of RBV theory

- Considered **static** in concept (Prime and Butler, 2001).
  - Not directly address the competitive implications of **market changes** as they relate to an organizational ability to **adapt its resources** and competencies to a rapidly changing environment. i.e., **durability** and/or **relevancy** challenges. – **flexibility & agility challenges** + **Lack of clarity in defining valuable resources**.
  - Challenges in explaining the differing competitive advantages seen across industries – **context dependency**.
- **The mechanisms to explain the linkage between resources and performance are not well identified** (e.g., Coff, 1999) “..the processes through which particular resources provide competitive advantage remain in a **black box**” (Barney 2001, p. 33).
  - **Limited attention to the mechanisms** by which resources are converted to competitive advantage (Eisenhardt and Martin, 2000).
  - Strategic resources contribute to performance only to the extent that organizations **capture the economic value they create** (Barney and Clark 2007 ). – **a business model context!**
- The RBV overemphasizes internal capabilities, neglecting external factors, and it is a retrospective tool that gives management limited practical use because it is **not able to explain the evolutionary nature of resources**. (Zahra and George, 2002).

# 4. Dynamic capabilities - The "2.0" of RBV

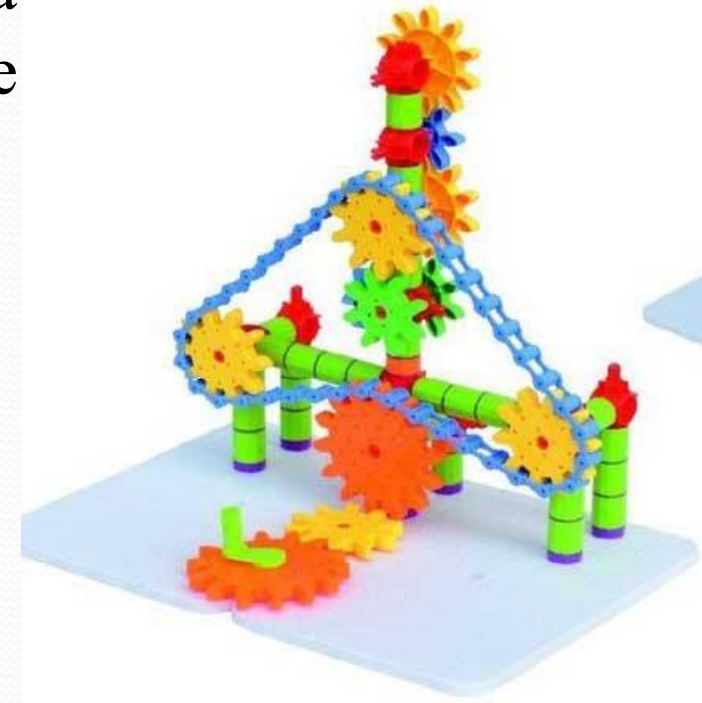
- Attempt to address the deficiencies of the RBV approach by looking into:
  - **The dynamic part** – is the ability to adapt a company's resources and competencies (both tangible and intangible) to a rapidly changing environment.
  - **Open the "black box"** - by **strengthening the link between resources and performance** by clarifying the **mechanisms** by which resources are converted to competitive advantage (not by converting intangible assets into numbers).
  - **Strategy-Operations fit** - Giving management more practical use in **linking strategic objectives, competencies, and capabilities** (both present and planned) **with operational objectives, competencies, and capabilities**, so that organizations could capture more successfully the economic value they plan to create.
  - **Better internal-external fit**. Closely tied to **learning to learn**, which enables organizations to continually innovate and adapt to changes.
- (Teece et al. 1997) "We define dynamic capabilities as the firm's ability to integrate, build, and reconfigure internal and external competencies to address rapidly changing environments".
  - Dynamic capabilities thus reflect an organization's ability to achieve **new and innovative forms of competitive advantage, regardless to present path dependencies and market positions** (Leonard-Barton, 1992)".
    - **Instead of closing the leaking holes in the bucket, add more "water" i.e. new capabilities.**

# Dynamic Capability theory is an **extension**, not cannibalization, of previous competitive paradigms

- **Traditional elements of business success in previous models** e.g., maintaining incentive alignment, owning tangible assets, controlling costs, maintaining quality, and optimizing inventories **are necessary but not sufficient for sustained superior performance in changing environments** (Helfat et al. 2007; Teece 2007 ).
- “Dynamic capabilities are more than a simple addition to the resource-based view since they manipulate the resources and capabilities that directly engender rents” (Zott, 2003: 120).
  - ..‘our view of the firm is somewhat richer than the standard resource-based view...it is not only the bundle of resources that matter, but the mechanisms by which firms learn and accumulate new skills and capabilities, and the forces that limit the rate and direction of this process.’ (Teece et al. 1997)

# Now, what do we mean by “Capabilities”?

- ..a capability is a **set of processes and activities** learned by a firm, which allows for **generating a specific result** (Teece et al. 1997).
- **A single Capability** is: ”The **capacity** to perform a particular activity in a **reliable** and at least minimally satisfactory manner” (Helfat and Winter, 2011, p. 1244).
- **Organizational capabilities** pay attention to the **deployment, combination, and synergistic linkages between single capabilities.**

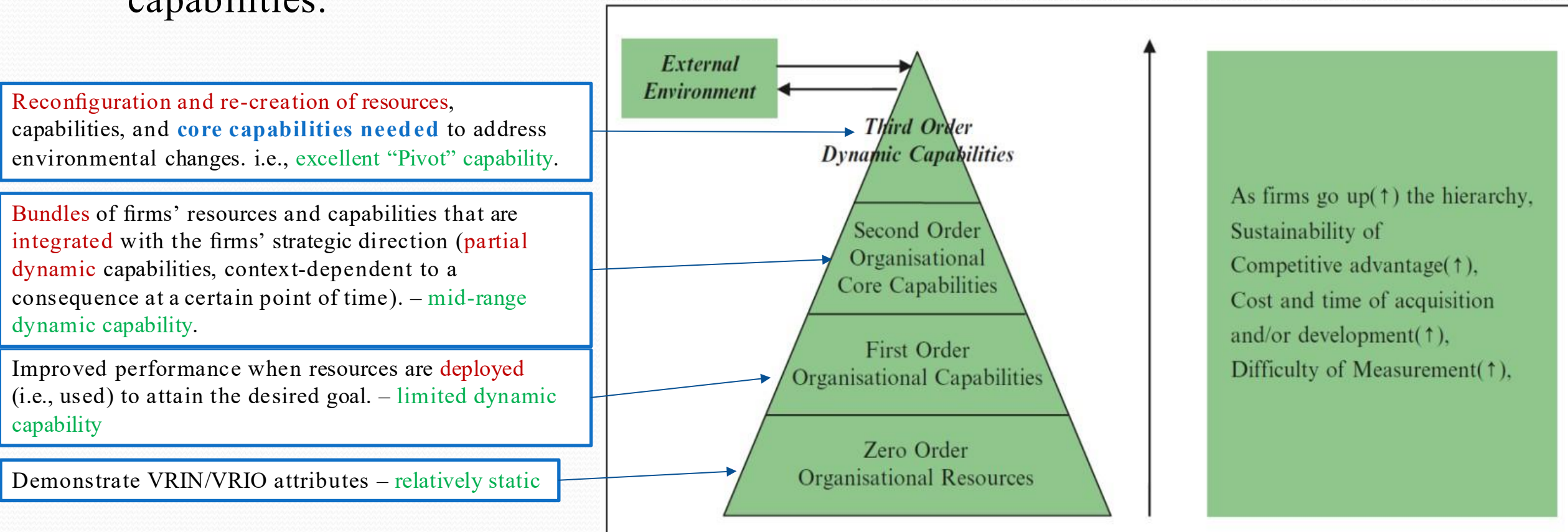


# Operationalizing organizational “Capabilities”

- Organizational capabilities involve:
  - **Ordinary capabilities** (operations, administration, governance) are necessary but not sufficient for long-run (financial) success – a process of developing efficient exploitation and optimization capabilities.
  - **Dynamic capabilities** – are both necessary and needed for (short and) long-run financial success. Often involves a **change in strategy** or in **business models**, and **calls for a good exploitation-exploration balance, AND flexibility & agile capabilities**.
    - E.g., New product and process development capability; R&D, IPR technology transfer capability; Market disruptiveness capability; Flexible and **agile** manufacturing capabilities; Human resource management capabilities, learning capabilities, etc.
  - **Capabilities are not synonymous with resources:**
    - **Capabilities** represent a firm’s **unique ability** or **capacity to identify** distinct internal and external competencies and subsequently coordinate, **reorganize and reshape them** as environmental changes dictate or allow.
    - **Resources** can be externally acquired and directly infused into an organization. (Jiang, 2014).

# Today's operational capabilities vs. tomorrow's dynamic capabilities

- A four-level hierarchy proposed by Wang and Ahmed (2007) is useful to understand the concepts of firm resources, capabilities, and dynamic capabilities.



# Flexibility & Agility

- **Agility** - the ability of an organization to **respond quickly and effectively to environmental changes**.
- (organisational) **flexibility** - the ability to **adjust processes, structures, and resources when conditions changes**.
- While **flexibility** enables adaptation at the operational level - involves altering existing procedures and organizational arrangements when circumstances require it.
- **Agility** reflects a broader organizational capability combining flexibility, speed, and responsiveness - it describes the firm's broader ability to detect changes in the environment and react swiftly and effectively.
- An **agile approach to innovation** (linked to dynamic capabilities) involves developing and testing ideas in **small, iterative stages** instead of trying to create developed solutions from the start. This approach can facilitate the early detection and resolution of issues and allow for necessary adjustments to be made along the way.

# Sensing, Seizing and Reconfigure

(Teece, 2008)

- Dynamic capability is **a set of actions or routines** taken by senior management that permits the firm to identify or sense (and shape) opportunities and threats, formulate a response to such opportunities and threats, and reconfigure assets to adapt to these (Helfat et al. 2007; Katkalo et al. 2010; O'Reilly and Tushman 2007; Teece 2008 ). **CHANGE IS THE ROUTINE!**
- Teece – with reference to DC's, what matters for business is: "the capacity 1) to **sense** and shape opportunities and threats, 2) to **Seize** opportunities, and (3) to maintain competitiveness through enhancing, combining, protecting, and when necessary, **reconfiguring** the business enterprise's intangible and tangible assets"
- Helfat et al. ( 2007 ) used the term '**asset orchestration**' to include these three functions of dynamic capabilities: sensing, seizing, and reconfiguring. They are the micro-foundations (Teece 2007 ) or the enablers (Ambrosini and Bowman 2009 ) of dynamic capabilities.

# Sensing



- Sensing and shaping new opportunities requires firms to constantly **scan, create, search, learn and interpret across technology and market**, locally and distantly (Teece 2008 ). **Opportunities can be detected in two ways: differential access to new information and/or new technology.**
- A set of resources and routines are involved, such as investments in:
  - **[Internal] Research activity**; probing and re-probing of customer needs and technological possibilities; internal discussion and argument of new opportunities; understanding latent demand; the structural evolution of industries and markets (Teece 2008 ).
  - **[open] Information search** should embrace the whole business ecosystem, including all potential active collaborators (customers, suppliers, and complementors) both locally and distantly (external).

# Seizing



- Once opportunities and threats are sensed, firms need to act on them in a **timely manner**, which is seizing challenges. Not surprisingly, some firms with new technology still failed to capture value from it, due to the **lack of seizing capability** (Chesbrough and Rosenbloom 2002 ).
- Seizing opportunities, which can be referred to as **strategic execution or strategic decision skills, is making the right decisions and executing them into new products, processes, or services** (Teece 2008). This capability requires a clear delineation of customer solutions **and a business model** that defines commercialization strategy and investment priorities.
- Developing seizing capability also needs a **decision-marking protocol** that recognizes complementarities, **avoids decision errors and biases**, as well as commits resources to the opportunities. **It is essential for incumbent managers to avoid the decision trap that is making decisions relying on path-dependent routines, assets, and developed strategies for existing technologies.**

# Reconfigure



- Firm growth and profits can be achieved **when new opportunities are sensed and effectively seized**. But to sustain the growth and profits, firms need the **ability to continuously recombine and reconfigure organizational assets and structures** as markets and technologies constantly change.
- The reconfiguration of existing resources and capabilities with new patterns of integration is **much more complex**, as it involves the **aligning of co-specialized assets to produce a new value**, thereby enhancing the combinations inside and between firms (Lockett et al. 2009; Teece 2008).
- Reconfiguration is the process or structure that redefines role systems and relational patterns in a flexible way, to make it easier to continuously **recombine resources** (Verona and Ravasi 2003 ).

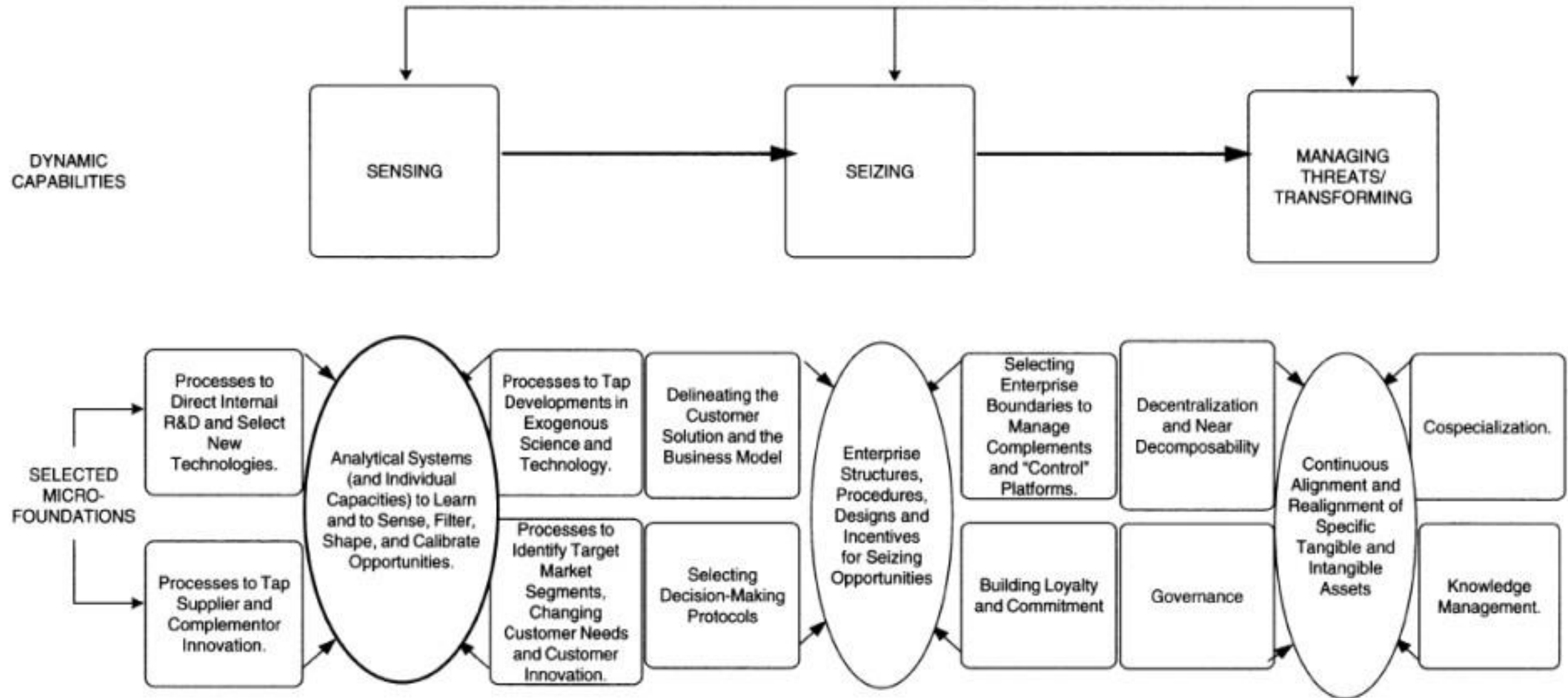
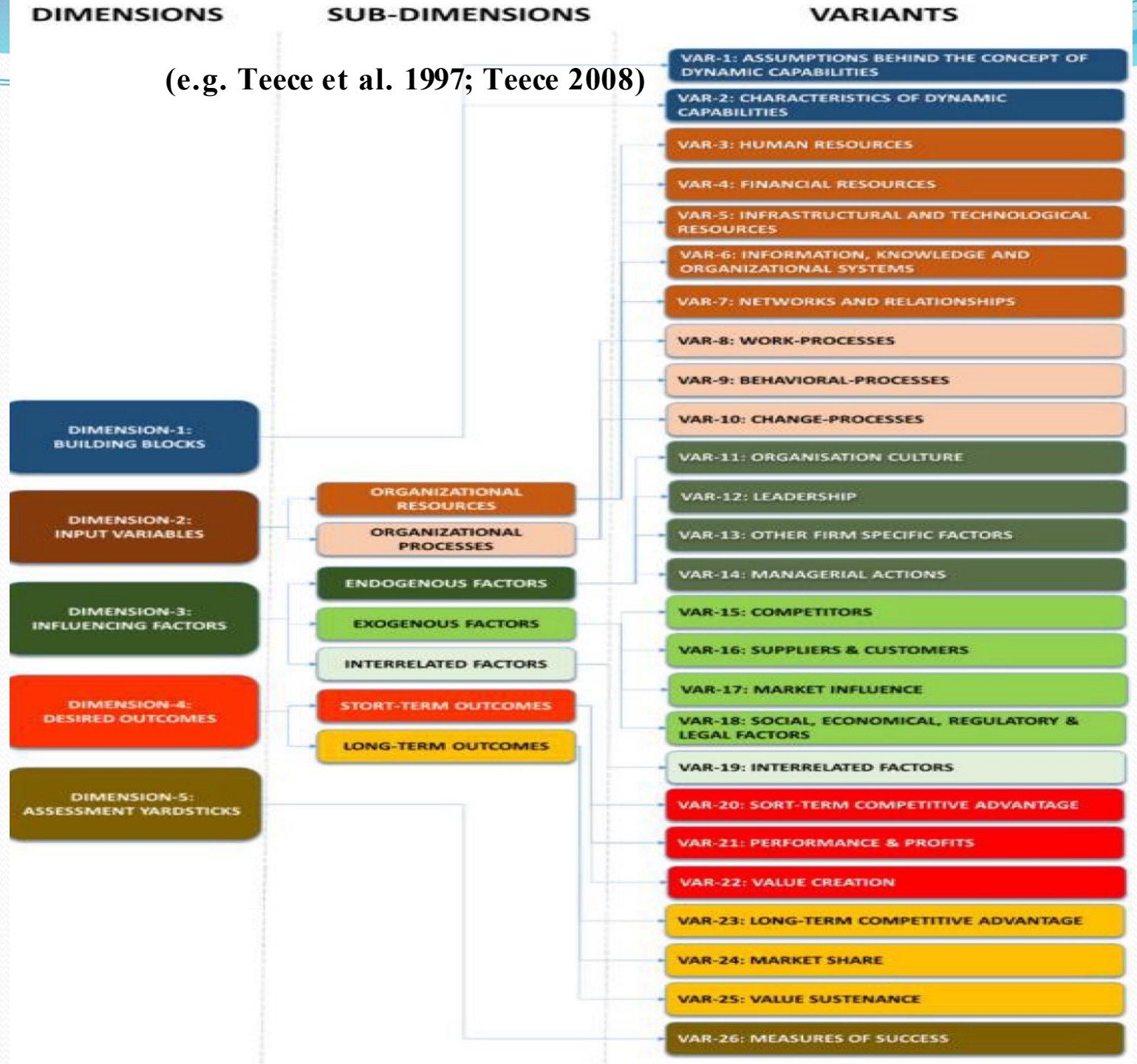


Figure 4. Foundations of dynamic capabilities and business performance

# Dynamic capabilities morphological analysis (MA) framework

Vijaya, Ganesh and Rahul (2019)



# Absorptive Capacity & Dynamic Capabilities

(Mazzarol and Reboud, 2020, p. 237)

- firm's (**realised**) ability to acquire, assimilate, transform and exploit information and knowledge represents an **absorptive capacity** that can **form the basis of a dynamic organizational capability** (Teece et al. 1997).
- The relationship between a firm's absorptive capacity and its ability to achieve success through dynamic capability is a function of **how well the management and their staff can allocate resources and use tacit and explicit knowledge to help them generate value through the bundling of resources into new combinations for commercial exploitation** (Gray 2006).

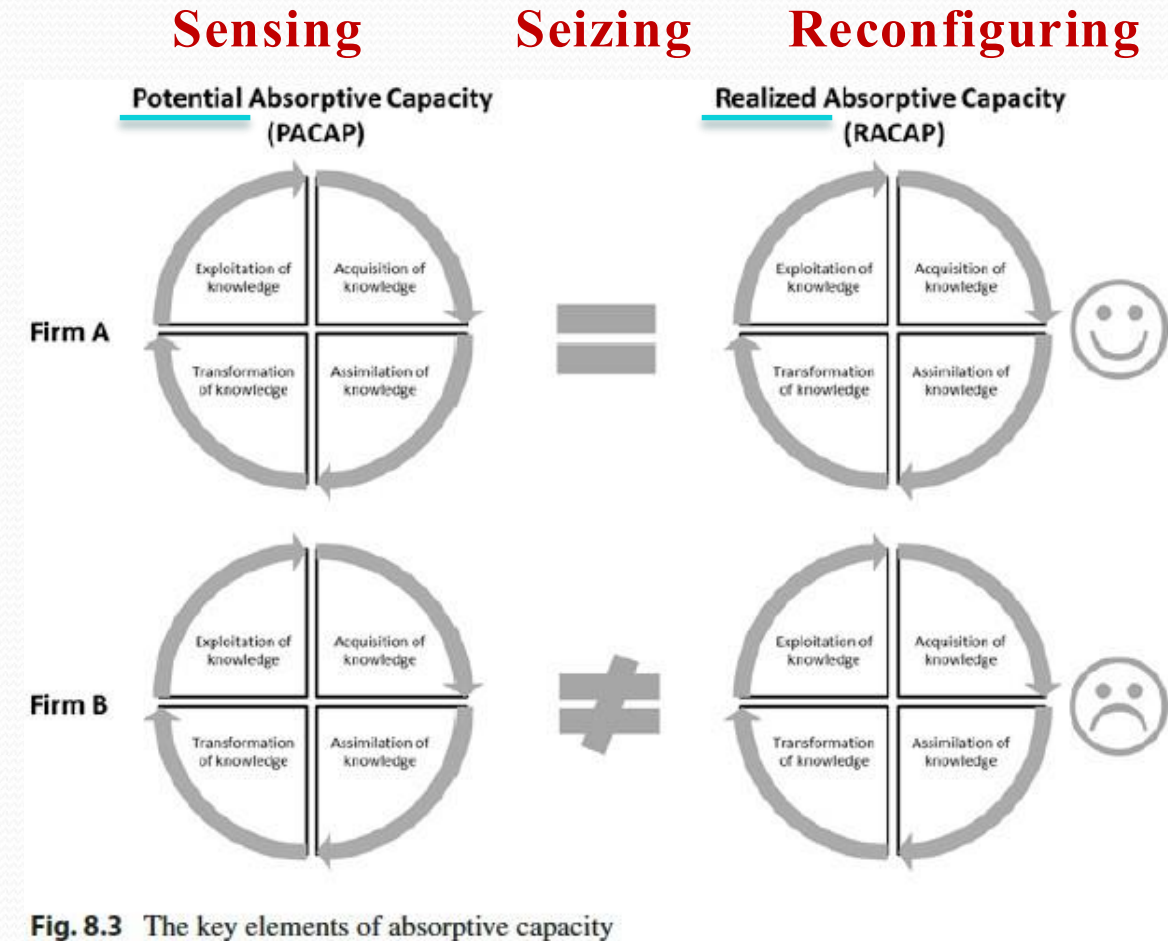


Fig. 8.3 The key elements of absorptive capacity

Firm A can successfully **realise** its potential absorptive capacity to acquire, assimilate, transform and exploit knowledge (good DC's), while Firm B cannot (..also risk management failure-related).

# Critique of dynamic capabilities

- **Operationalization challenges** - There is a growing demand to focus on **the actual process of developing and managing adequately DCs**, rather than developing further **abstract definitions** of DCs.
- **Scale and scope challenges:** When senior managers are confronted with the task of building DC, **they need to consider sometimes drastic fluctuations in the threshold capability definition standards** (e.g. **how far do we go? how do we manage the risks? Do we gamble or manage the process?**), making it more and more complex for companies to **understand the minimum requirements to remain in the game as an industry player**.
- DC are necessary but not sufficient for sustainable competitive advantage.
- **Food for thought: Can business model innovation theory help here?**

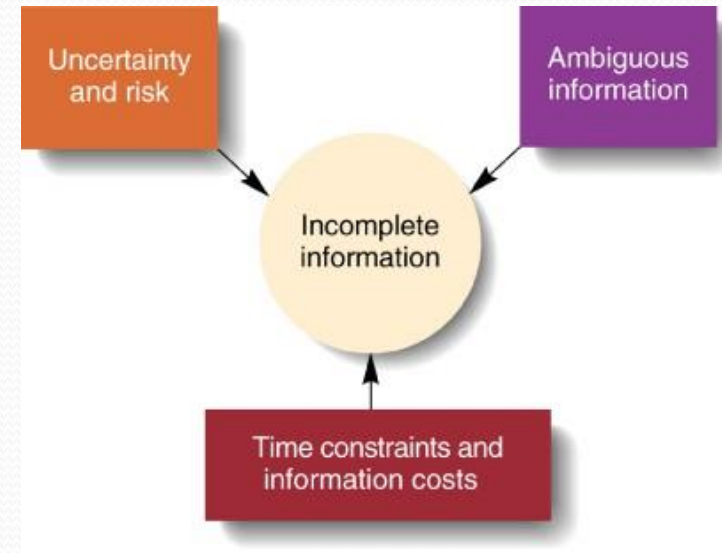
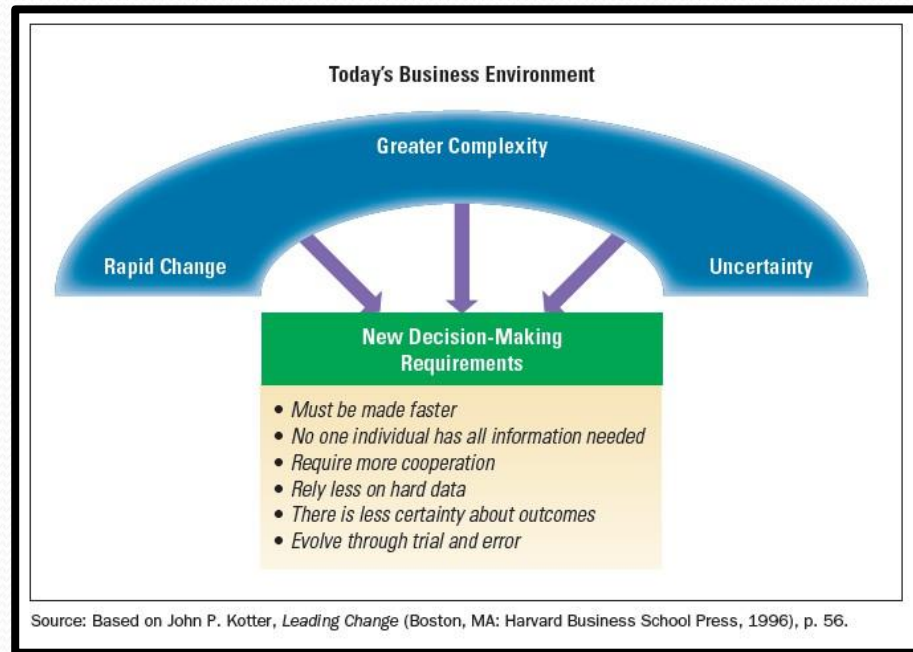
# Risk Management

- “By its very nature, innovation is inherently risky. The more radical and disruptive the innovation, the more uncertainty and potential risk is created. However, the management of risk remains an important issue for any manager or organisation seeking to engage in the commercialisation of innovation.” (Mazzarol and Reboud, 2020, p. 227)

BUT,

- “It is not necessarily subject to a specific investment and its impact on performance remains difficult to demonstrate, so risk management in new ventures has so far not been considered either as a source of specific cost, or as a source of significant value creation.” (Mazzarol and Reboud, 2020, p. 228)

# Bounded Rationality as an inherent outcome in today's "VUCA" business environment



# The problem: Do managers **gamble** or **manage** their decision-making choices?

- Managers do not have the time or money to search for all alternatives.
- **In effect:** At best, managers explore a limited number of options and choose an acceptable decision rather than the optimum decision.



<http://www.starresults.com/>

# Definitions

- **Risk** - “...the uncertainty of outcome” (Chapman and Ward 2004), or the “combination of the probability of an event and its consequences” (ISO/IEC Guide 73, 2002).
- **Risk Management** - “the systematic application of management policies, procedures and practices to the tasks of communicating, consulting, establishing the context, identifying, analysing, evaluating, treating, monitoring and reviewing risk” (ISO/IEC Guide 73).

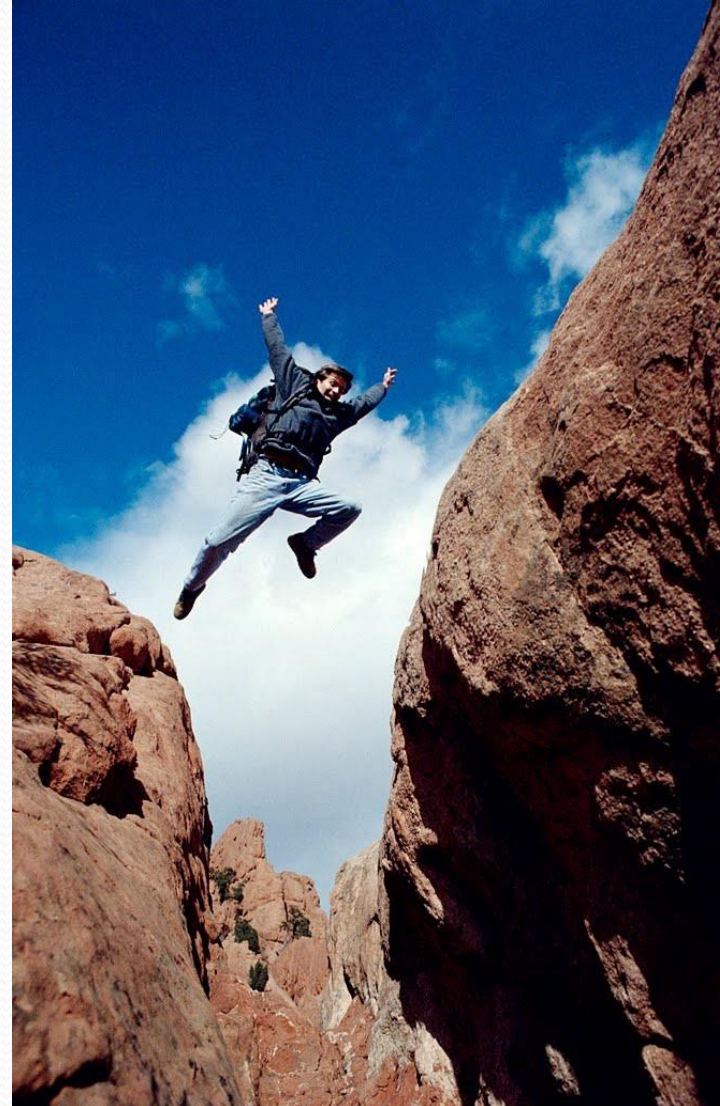
# Risk vs. uncertainty

- **Risk** is something that **can be measured in a quantitative [or qualitative] manner and assessed**. However, **uncertainty** is not so readily quantitatively measured. (Knight 1921)
- “Uncertainty is a condition caused by having insufficient information or knowledge to make accurate predictions about the likely outcome or future state of an action or event. **In other words, if you can determine the odds of something happening, you can assess the risk of gain or loss and make a rational choice. However, where uncertainty exists you cannot assess the risk so clearly.**” (Mazzarol and Reboud, 2020, p. 231)

# What's your risk appetite?

Put simply, risk appetite is “the total impact of risk an organization is prepared to accept in the pursuit of its strategic objectives” (KPMG 2008, p. 3).

Once the risk appetite threshold has been breached, RM treatments and business controls are implemented to bring the exposure level back within the accepted range.



# Three RM models that a company can adopt:

(Accenture, 2009)

1. RM for **compliance** where RM involves a regulatory set of requirements focused on keeping the organization compliance (e.g. with regulations).
2. RM for **value protection**, which is aimed at managing expected risks, as well as reducing the degree of unforeseen risks.
3. RM for **value enhancement**, which is aimed at covering all dimensions of the business, as well as increasing the protection against unforeseen risks.

# Today's problem 1/2

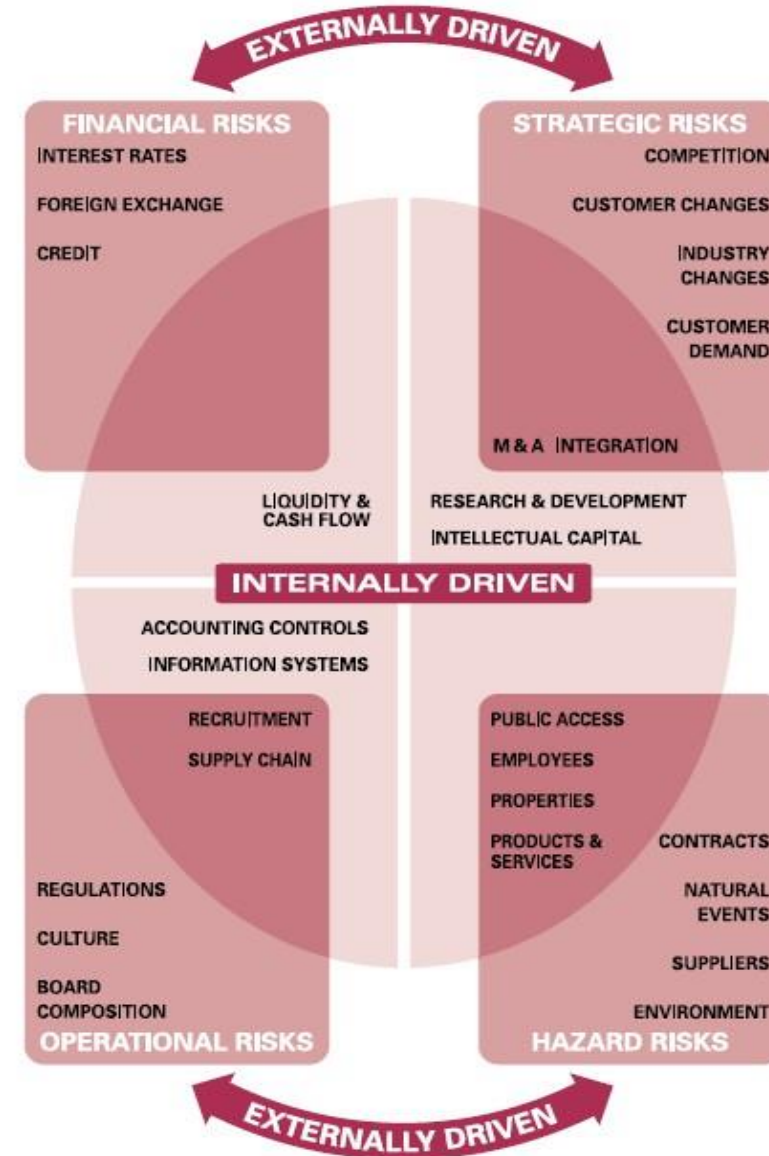
- RM processes have been successfully incorporated into few organizational units and/or industries such as **internal audit, treasury, insurance, environmental health & safety, and legal reports** . But, it has not yet been fully incorporated into other core business processes, such as **strategic planning, capital allocation, and performance management** (ERM survey, 2008).
- Consequently, risks are being managed mostly in a **reactive manner** (i.e. comply with regulations and/or value protection).

# Today's problem 2/2

- Yet, surveys (e.g. ERM survey, 2008) have shown that **a growing percentage of managers worldwide are showing a growing interest in applying risk management** in a much more comprehensive (i.e., **proactive and holistic**) manner.
- But ... How???

# Enterprise Risk Management (ERM)

- According to the Committee of Sponsoring Organizations (COSO), ERM deals with **risks and opportunities affecting value creation**, and helps an entity to get where it wants to go and avoid pitfalls and surprises along the way.
- Organizational risks can be classified into e.g., five main risk **groups**: Strategic risks, Operational risks, Financial risks, Cultural risks, and (external) Hazard risks.

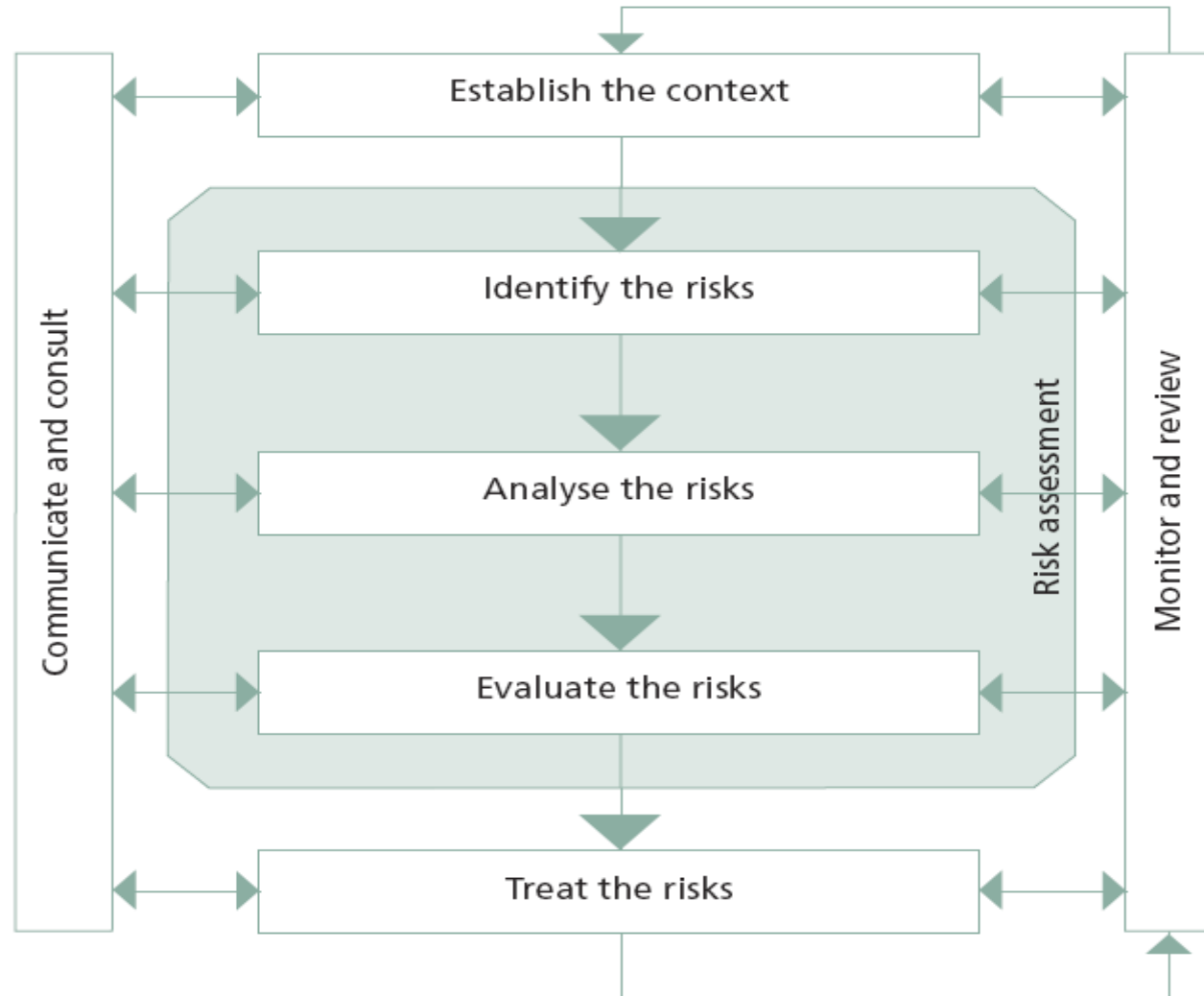


# The process of managing risks

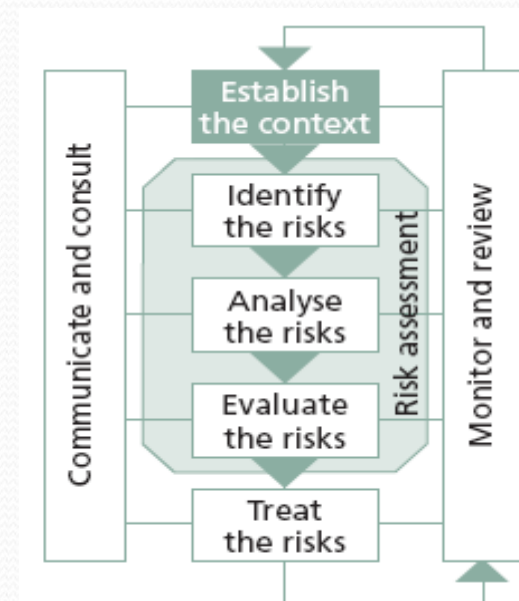
(Australian standard (AS 4360) 'Risk Management')

(Mazzarol and Reboud, 2020, p. 242)

Figure 3.1 The risk management process (AS/NZS 4360)

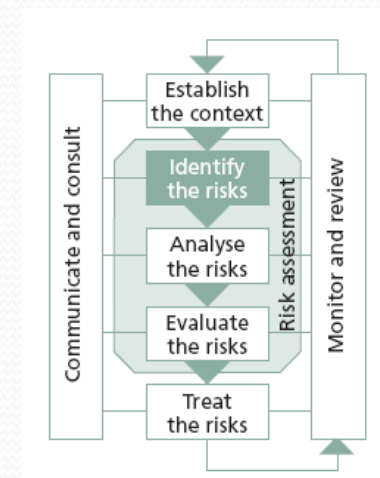


- **Communicate and consult** – identify who should be involved in assessment of risk (follow through the entire process).
- **Establish the context** – both **internal** (e.g. objectives and goals of a business) and **external** (e.g. environment in which a business operates) context.

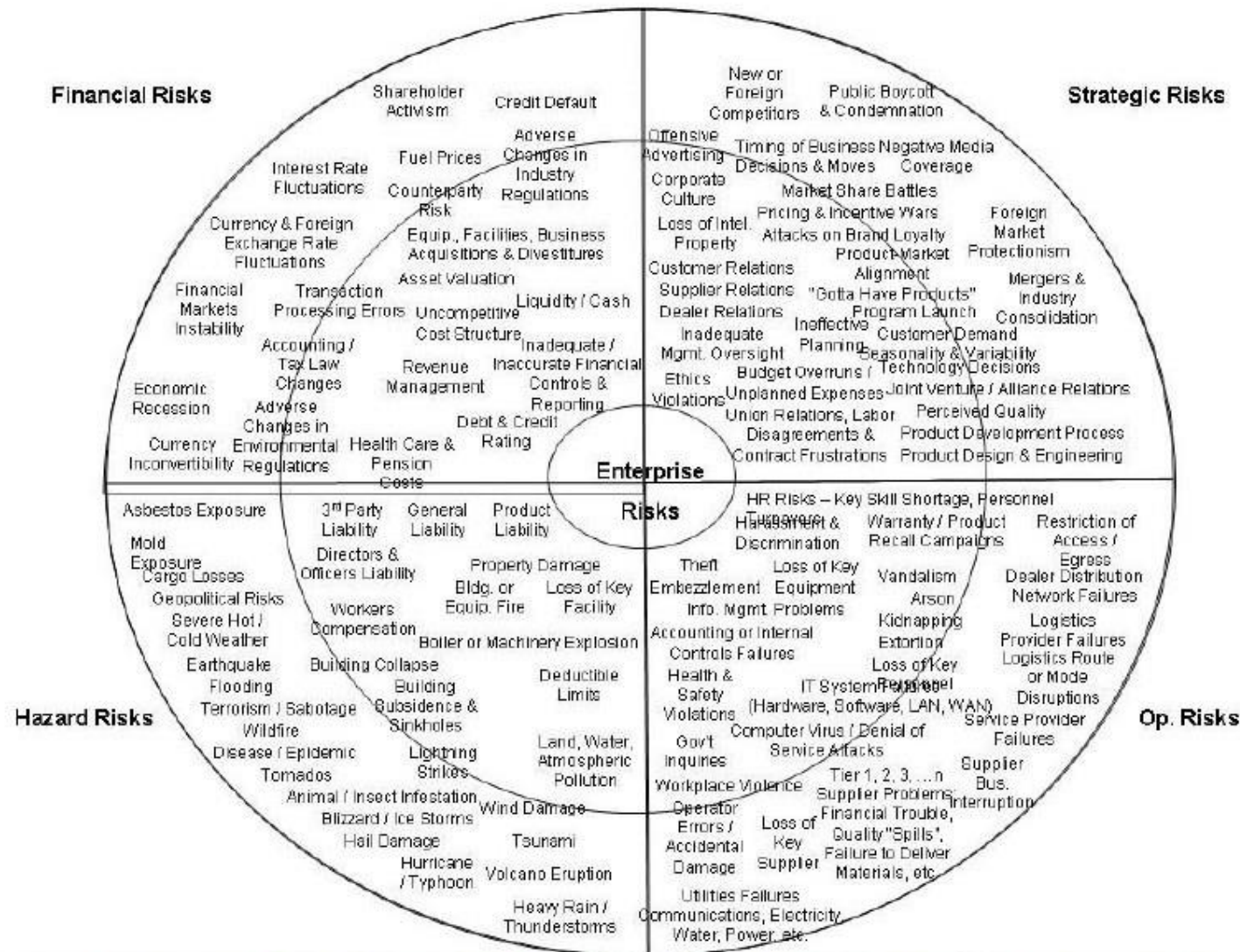


# Identify the Risks:

- Today, in most companies who apply RM programs, risks are being identified by executive management team, individual operating teams, project teams and the various Board sub-committees (The national affordable homes agency, 2008).
- **What can happen? How can it happen? Why could it happen?**
- a risk factor portfolio can be developed:
  - Retrospectively
  - Prospectively (e.g. learning from others' mistakes)



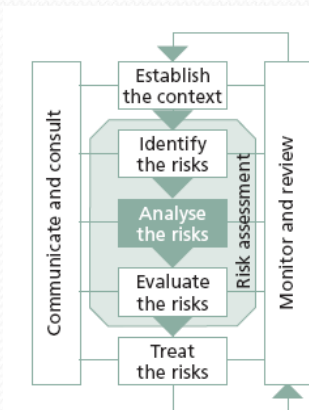
# Portfolio of risks - example



Source: Debra Elkins, "Managing Enterprise Risks in Global Automatic Manufacturing Operations,"

# Analyzing and evaluating risks 1/2

- **Inherent risk assessment** - [The **likelihood** of something happening] x [The **impact** if it does happen]. For each risk identified, an **inherent risk** assessment is made, where:
  - “**Low**” risks (green color), are considered being an acceptable level of risk exposure, and therefore no further action is needed.
  - “**Medium**” (orange color) and especially “**High**” (red color) risks, are considered being an unacceptable level of risk exposure, and requires further action

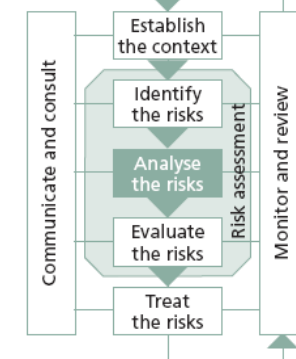


# Analyzing and evaluating risks 2/2

## Risk Matrix

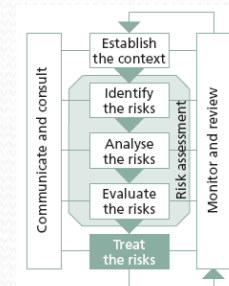
Example of inherent risk assessment process:

| Impact        | Multiplier |      |          |          |        |                |
|---------------|------------|------|----------|----------|--------|----------------|
| Catastrophic  | 5          | 5    | 10       | 15       | 20     | 25             |
| Major         | 4          | 4    | 8        | 12       | 16     | 20             |
| Moderate      | 3          | 3    | 6        | 9        | 12     | 15             |
| Minor         | 2          | 2    | 4        | 6        | 8      | 10             |
| Insignificant | 1          | 1    | 2        | 3        | 4      | 5              |
|               |            |      |          |          |        |                |
|               | Multiplier | 1    | 2        | 3        | 4      | 5              |
| Likelihood    |            | Rare | Unlikely | Possibly | Likely | Almost Certain |



# Treating risks

- From risk management perspective, for each risk who was categorize as “Medium” or “High”, the treating process choices should be narrowed into the following possibilities (e.g. DeLoach, 2003):
  - **Avoiding** (eliminating) the risk.
  - **Reducing** the risk.
  - **Accepting** the risk.
  - **Transferring** or **sharing** the risk.
- Clearly, each risk brings with it a different treating process. Some treating processes can be tailored specifically to **individual risks** (e.g. action to reduce the risk), while others could treat **cluster of risks** simultaneously (e.g. share/transfer).
- **Monitor and review** - review the effectiveness of the treatment plan (last phase).



# Example of a risk management process (Taran, Boer and Lindgren, 2013)

| Qualitative risk analysis                                                                |                  |              |                                  | Action plan                                                                                                                                                                                      |                                                                                                                                             | Post-hoc evaluation             |                                    |                                                                                 |
|------------------------------------------------------------------------------------------|------------------|--------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------|---------------------------------------------------------------------------------|
| Medium and high risks identified                                                         | Likelihood (1-5) | Impact (1-5) | Inherent risk (before treatment) | Risk treatment                                                                                                                                                                                   | Person in charge, and milestones                                                                                                            | Residual risk (after treatment) | Fit to the company's risk appetite | Further action planned                                                          |
| <b>Strategic risks</b>                                                                   |                  |              |                                  |                                                                                                                                                                                                  |                                                                                                                                             |                                 |                                    |                                                                                 |
| S1 – Provital's lack of sales, leads to a shutdown of the joint venture.                 | 5                | 5            | High                             | Primary focus is on insuring the company owners' satisfaction through improving web visualization, aggressive marketing, achieving immediate sales and revenues, and securing IPR on the system. | Top manager and R&D manager are responsible, already working on solving the problem. They continue more rigorously after receiving funding. | Low                             | Fit                                | No further action needed. But keep monitoring closely the owners' expectations. |
| S2 - New competitor enters the industry with a competitive solution.                     | 4                | 3            | Medium                           | Monitoring the industry for potential competitors.                                                                                                                                               | R&D manager                                                                                                                                 | Medium                          | No fit                             | Keep monitoring the industry for potential competitors on a monthly basis.      |
| <b>Operational &amp; cultural risks</b>                                                  |                  |              |                                  |                                                                                                                                                                                                  |                                                                                                                                             |                                 |                                    |                                                                                 |
| O1 - No professional sales people. Low marketing skills and practice.                    | 1                | 5            | Medium                           | Need to get funding for employing high quality sales and marketing people. Also, considering outsourcing the marketing function and selling to a third (expert) company.                         | All company managers. Partly already in progress, and to be applied more rigorously after receiving funding.                                | Low                             | Fit                                | No further action is needed.                                                    |
| O2 - Tests fail to show that the system is successful also in other industrial settings. | 2                | 3            | Medium                           | Insure that the system operates successfully before sales. The system will not sell if the pre-stress tests show that the system fails to operate successfully.                                  | R&D manager. Already working on the problem.                                                                                                | Low to medium                   | Poor fit                           | No further action needed.                                                       |
| O3 - One of Provital's suppliers choose to stop working with the company.                | 1                | 1            | Low (many suppliers available)   | -                                                                                                                                                                                                | -                                                                                                                                           | -                               | -                                  | No further action needed                                                        |
| <b>Financial risks</b>                                                                   |                  |              |                                  |                                                                                                                                                                                                  |                                                                                                                                             |                                 |                                    |                                                                                 |
| F1 - Lack of investment money.                                                           | 3                | 4            | Medium                           | Looking for potential investor.                                                                                                                                                                  | All managers (and owners). Already working it.                                                                                              | Low to medium                   | Poor fit                           | No further action needed.                                                       |
| F2 - One of the mother companies goes bankrupt.                                          | 1                | 5            | Medium                           | Cannot be controlled by the company.                                                                                                                                                             | -                                                                                                                                           | Medium                          | No fit                             | Tolerate.                                                                       |
| <b>Hazard risks</b>                                                                      |                  |              |                                  |                                                                                                                                                                                                  |                                                                                                                                             |                                 |                                    |                                                                                 |
| H2 - Global financial crisis affects the company's performance and sales.                | 5                | 3            | High                             | Each financial investment will be carefully analyzed and decided on jointly with financial experts.                                                                                              | Top manager and R&D manager are responsible. Already in progress.                                                                           | Low to medium                   | Poor fit                           | No further action needed.                                                       |

# Risk management should be:

- Considered as a **central hub** – risk network.
- **[a] Decentralized** program – everyone should be involved.
- **Shared and communicated** in, and between, departments, and to other relevant stakeholders. Both in identifying and in treating relevant risks.



**Thank you!**